1. INTRODUCTION

Menstrual health is a vital component of women's overall well-being, yet it often remains under-discussed and under-addressed in healthcare systems. Accurate cycle tracking is essential not only for fertility awareness but also for early detection of irregularities that may indicate underlying health concerns. With the widespread use of smartphones, digital health solutions have become popular tools for women to monitor their cycles and related symptoms. However, most available applications rely on static, calendar-based predictions, limiting their reliability when cycles are irregular. The rise of Artificial Intelligence (AI) in healthcare presents an opportunity to revolutionize menstrual tracking by making it more personalized, adaptive, and insightful.

1.1 Background / Context:

Earlier, women relied on manual calendar tracking, which could not handle irregularities or provide health insights. Modern apps improve tracking but still depend on static calculations and often fail in cases of irregular cycles, inconsistent symptoms, or sudden health changes.AI has influenced healthcare by offering pattern recognition, time-series prediction, anomaly detection, and personalized recommendations. Applying AI to menstrual tracking enables a system that learns from user data—period dates, symptoms, mood patterns, lifestyle habits—and improves prediction accuracy overtime.Therefore, AI-based cycle tracking is capable of overcoming the limitations of traditional methods by offering adaptive, personalized reproductive-health insights.

1.2 Problem Definition:

Existing menstrual trackers face the following problems:

- Predictions are inaccurate for irregular cycles.
- Apps use static formulas, not learning models.
- No early identification of abnormal patterns.
- Symptoms and lifestyle data are rarely analyzed.

- Poor adaptation to long-term changes.
- Privacy issues in commercial apps.

This project aims to solve these problems by creating an AI-powered tracker that provides personalized cycle predictions, symptom-based insights, and early-warning alerts.

1.3 Objectives:

The main objectives of the AI Integrated Period Tracker are:

- To predict period start dates, ovulation windows, and fertility phases using AI/machine learning.
- To analyze symptoms such as cramps, mood swings, flow pattern, sleep issues, and stress.
- To identify cycle irregularities early and notify the user.
- To provide personalized health insights and reminders.
- To design an intuitive interface for easy tracking.
- To ensure secure storage and privacy of user health data.

1.4 Scope of the Project:

Phase-I:

Requirement analysis, literature review, system analysis, system design, initial dataset preparation, implementation of period-logging and symptom-logging modules, preprocessing of menstrual cycle data, integration of base AI prediction models, and creation of a partial prototype for cycle prediction.

Phase-II:

Full AI model training using real or expanded datasets, complete frontend–backend integration, UI/UX development for the mobile application, implementation of ovulation and fertility prediction modules, addition of lifestyle correlation features,

multilingual support, notifications and reminders, system testing, optimization, and final deployment of the AI Integrated Period Tracker.

1.5 Significance / Need of the Study:

- Women often struggle to track irregular periods, symptoms, and hormonal changes manually.
- Existing apps use simple averages and fail to provide accurate, personalized predictions.
- There is a need for an intelligent system that analyzes symptoms, lifestyle factors, and patterns to improve menstrual health awareness.
- An AI-based tracker can provide accurate predictions, early alerts, and better insights, reducing effort and improving health monitoring.

1.6 Methodology Overview:

The methodology for the project includes:

1. Data Collection: Period dates, symptoms, lifestyle factors.
2. Preprocessing: Cleaning missing entries, encoding symptoms, normalizing cycle lengths.
3. Model Development: Using LSTM (Long Short-Term Memory), Random Forest Regression, or time-series models.
4. Training & Testing: Splitting data into training/testing sets and evaluating predictions.
5. System Design: Architecture creation, UML diagrams, UI mockups.
6. Partial Implementation: Creating basic UI and integrating initial prediction functions.
7. Validation: Comparing model output with expected cycle patterns.

2. LITERATURE REVIEW

2.1 Past Research Work:

Research Work	Year	Key Findings	Limitations
Period Tracker Apps (Flo, Clue)	2020	Track menstrual dates and symptoms	Lack emotional support
AI in Women's Health	2021	Machine learning improves prediction accuracy	Lacks personalization
Chatbots for Healthcare	2022	Interactive query resolution improves accessibility	Limited reasoning and personalization
Mood-based Recommender Systems	2023	Suggests entertainment based on emotion	Not connected with menstrual data

Table no1: Comparative Table of Prior Research Works

2.2 Existing Systems Summary:

Existing menstrual-tracking solutions include:

- Calendar-based apps (static prediction, average calculations)
- Symptom loggers (manual entry with no predictive analysis)
- Fertility prediction apps (limited dataset integration)

Limitations of existing systems:

- Most do not use AI for adaptive predictions
- Limited analysis of symptoms, mood patterns, or lifestyle data
- No early detection of irregularities
- Minimal personalization,Data privacy concerns

2.3 Comparison of Approaches:

Feature / Criteria	🩸 Existing Applications (e.g., Flo, Clue, My Calendar)	🤖 Proposed AI Integrated Period Tracker
Cycle Tracking	Basic tracking using fixed 28-day or user-input cycles	AI-based prediction using **Machine Learning (TensorFlow)**
User Interaction	Limited pre-defined responses or FAQs	Smart **Chatbot (GPT API)** for personalized guidance
Mood Tracking	Manual input of emotions	**Emotion detection model (Hugging Face)** for automatic mood understanding
Food Recommendations	Generic diet tips	**AI-generated personalized food suggestions** based on mood & cycle phase
Entertainment Support	Rare or not included	**Mood-based games, music, and videos** for emotional balance
Notifications	Basic period reminders	**Firebase Cloud Messaging** for intelligent reminders and wellness tips
Data Storage	Local or third-party cloud	Secure **Firebase/MongoDB database** for user records
Technology Stack	Basic mobile app frameworks	Modern stack: **Python, React.js, TensorFlow, FastAPI, GPT API**
Goal	Period tracking only	**Holistic women's health support** (tracking + mood + food + wellness)

Table no. 2: Existing VS New System

2.4 Identified Research Gaps:

- Most tracking apps rely on average calculations instead of advanced AI-based analytics.
- Current systems fail to integrate lifestyle factors like stress, sleep, diet, or emotional state.
- Irregular cycles (common in PCOS, hormonal imbalance) are not handled well by traditional models.
- Existing tools do not support early anomaly detection or health alerts.
- Limited research models integrate symptoms, flow patterns, and mood variations into predictions

3. SYSTEM ANALYSIS

3.1 Existing System:

Traditional apps predict dates using simple averages and do not analyze long-term patterns, irregularities, symptoms, or lifestyle influences.

3.2 Limitations of Existing System:

- Inaccurate with irregular cycles
- No AI-driven learning
- Lack of personalized health insights
- Weak symptom-cycle analysis
- Limited predictive capability

3.3 Proposed System:

The proposed AI Integrated Period Tracker provides:

- AI-based cycle predictions
- Symptom analysis
- Pattern detection
- Personalized insights
- Secure user data handling

3.4 Feasibility Study:

Technical Feasibility:

- Python, TensorFlow, Firebase, Flutter can be used.
- ML models for prediction are available.
- System is technically possible.

Operational Feasibility:

- Easy to use and Suitable for adolescent girls and women of reproductive age.

Economic Feasibility:

- Most tools are open-source.
- Low development cost.
- Affordable for academic project

4. SYSTEM REQUIREMENTS

4.1 Functional Requirements

•User Data Ingestion:

System must allow users to log period dates, symptoms, mood, and lifestyle details.

•Cycle History Management:

System must store, retrieve, and update menstrual cycle logs for each user.

•Data Preprocessing:

System must clean missing entries, normalize cycle lengths, and encode symptom data.

•Feature Extraction:

System must compute cycle length variations, symptom patterns, and lifestyle correlations.

•AI Prediction:

System must output predictions for next period date, ovulation window, and fertile phase.

•Pattern Detection:

System must identify irregularities such as cycle delay, abnormal symptoms, or flow inconsistencies.

•Notification Handling:

System must send reminders for upcoming periods, ovulation, and health alerts.

•Data Visualization:

System must display charts for cycle history, flow pattern, and symptom trends.

4.2 Non-functional Requirements

- **Accuracy:**

  AI model should achieve high prediction accuracy for period dates and cycle patterns (Target: prediction error < 10%).

- **Reliability:**

  System must handle incomplete or inconsistent user data without crashing.

- **Scalability:**

  System must support multiple users and increasing cycle logs without performance issues.

- **Usability:**

  UI should be simple, intuitive, and mobile-friendly for daily usage.

- **Security:**

  Sensitive menstrual data must be encrypted and protected with secure authentication.

- **Performance:**

  Predictions and chart loading should occur within 2–3 seconds.

- **Compatibility:**

  System should run smoothly across Android, iOS, and Web platforms (if deployed).

4.3 Hardware Requirements

- **Processor:**

  Intel Core i5 / AMD Ryzen 5 or higher (for model training). For mobile users: Any modern smartphone (Android/iOS).

- **RAM:**

  8 GB minimum (16 GB recommended for faster model training).

- **Storage:**

  256 GB SSD recommended; user app requires < 200 MB.

- **Optional Sensors:**

  - Smartphone temperature sensor (if integrated)
  - Health tracking device / wearable

4.4 Software Requirements

- **Operating System:**
  - Windows,Linux,MAC /Android,IOS
- **Programming Languages:**
  - Python 3.10+ (for AI Model) Dart / Flutter (for mobile app)
- **AI / ML Libraries:**
  - TensorFlow / Keras
  - NumPy, Pandas, Scikit-learn
- **Backend Framework:**
  - Firebase / Django REST Framework / FastAPI
- **Database:**
  - Firebase Firestore / SQLite / PostgreSQL
- **Development Environment:**
  - VS Code
  - Jupyter Notebook
  - Android Studio (for app build)

5.SYSTEM DESIGN

5.1 System Architecture:

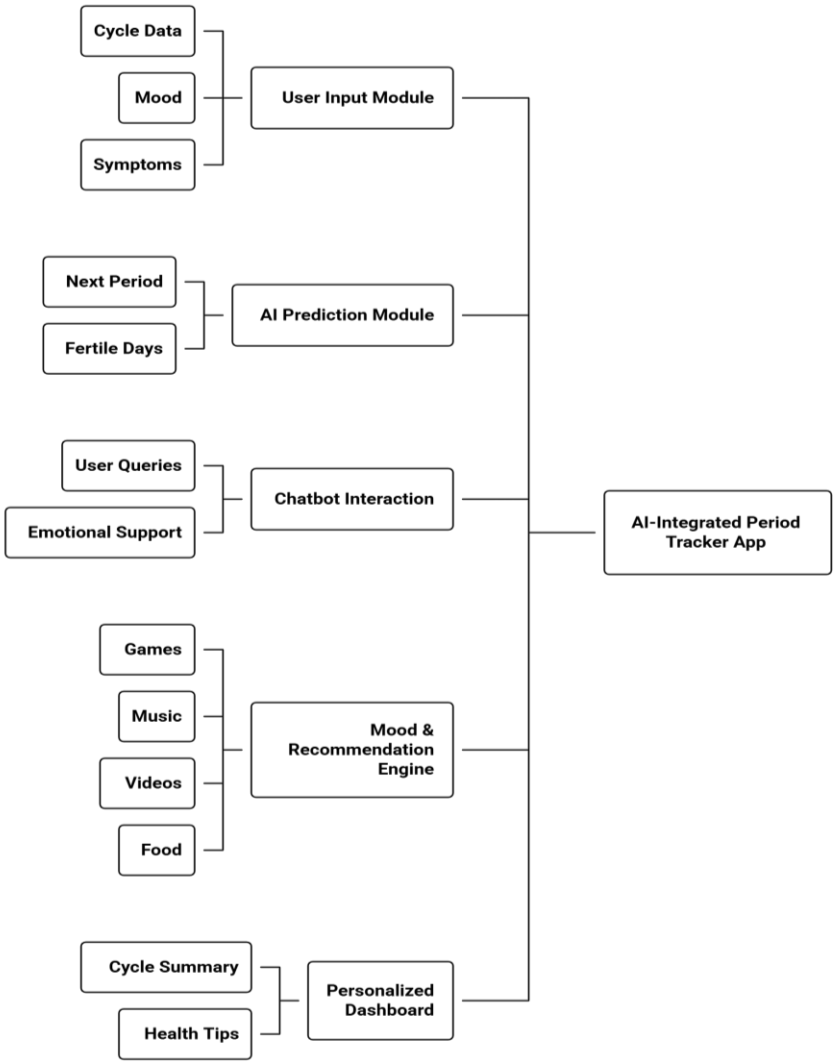


Fig.1: Overall System Architecture for AI integrated peroid tracker

1.User Module

The User Module is the primary point of interaction between the user and the application. It provides a clean, intuitive interface through which users can enter essential menstrual information such as period start date, end date, flow level, symptoms, and mood variations. This module is responsible for collecting accurate data that forms the foundation for all predictions and recommendations. A user-friendly interface ensures that logging data becomes a regular and effortless habit, which improves the accuracy of the entire system.

In addition to data entry, the User Module also helps users navigate through different features of the application, including viewing their cycle history, checking predictions, updating personal settings, and accessing helpful resources. This module is designed with simplicity and comfort in mind so that users feel confident while interacting with the app. Since the module acts as the bridge between the user and the system's internal mechanisms, it plays a crucial role in determining the overall user experience. The consistency and clarity of this module ensure that users can rely on the app for regular tracking, making it the foundation of the system.

2. AI Prediction Module

The AI Prediction Module serves as the intelligent core of the period tracking system. This module utilizes machine learning algorithms and statistical models to analyze historical user data, identify patterns, and predict upcoming menstrual events. By studying previous cycles, symptoms, lifestyle factors, and mood patterns, the AI is able to calculate expected future occurrences such as the next period date, ovulation day, fertile window, PMS timeline, and mood fluctuations. These predictions help users prepare mentally and physically for different stages of their cycle.

The AI model continuously learns and adapts as more data is added by the user. This continuous training improves prediction accuracy over time, providing personalized insights tailored to the individual's unique cycle. The module also identifies irregularities or abnormalities in user patterns, allowing the system to highlight potential areas of concern. Overall, the AI Prediction Module transforms raw data into meaningful, scientific insights that enhance the user's awareness and understanding of their reproductive health.

3. **Chatbot Interaction Module**

The Chatbot Interaction Module adds an interactive, conversational layer to the system, making it feel more supportive and user-friendly. This module allows users to communicate with the app through a chatbot that can answer their questions, clarify their doubts, and provide instant guidance. Users can ask questions related to their cycle, symptoms, predictions, or general menstrual health, and the chatbot responds in simple, understandable language. This reduces confusion and helps users feel more connected to the system.

The chatbot also assists in navigating the app by guiding users through tasks such as logging symptoms, checking predictions, or exploring various features. It sends reminders

to encourage daily logging and provides friendly motivation to improve user engagement. Additionally, the chatbot acts as a quick educational tool by explaining medical terms, cycle phases, and lifestyle tips in a supportive manner. This module ensures that users always receive instant, reliable assistance, making the app more interactive and accessible.

4.Mood & Recommendation System

The Mood and Recommendation System focuses on improving the user's emotional and physical well-being. By analyzing mood logs, symptoms, lifestyle habits, and AI predictions, this module generates personalized suggestions to support users throughout their menstrual cycle. It identifies emotional patterns and predicts days when users might experience stress, fatigue, irritability, or PMS-related mood dips. Based on this analysis, the system recommends actions such as healthy meals, relaxation techniques, hydration reminders, exercises, or self-care routines.

This module ensures that the support offered by the app extends beyond tracking and predicting periods. It provides daily or phase-based recommendations that help the user manage their health effectively. Whether it is suggesting a light walk for cramps, calming music during mood swings, or advising more water intake, the recommendations are timely and relevant. By combining emotional awareness with practical health tips, this module plays a vital role in improving the user's daily quality of life**.**

5.Personalized Dashboard Module

The Personalized Dashboard is the central display area where all important information is shown in a clear and organized manner. It visually presents cycle predictions such as expected period dates, ovulation window, fertile days, PMS days, and current cycle phase. The dashboard also displays mood patterns, symptom history, and lifestyle trends through charts or simple graphics. This allows users to quickly understand their health status at a glance.

The dashboard updates automatically as soon as new data is entered or new predictions are generated by the AI model. It helps the user monitor long-term changes and identify improvements or irregularities in their cycle over time. By offering a clean and personalized view, it becomes easier for users to track their health regularly and stay informed. The dashboard ties together all system insights into a single, easy-to-use space, making it an essential part of the user experience.

5.2 UML Diagrams: Sequence Diagram:

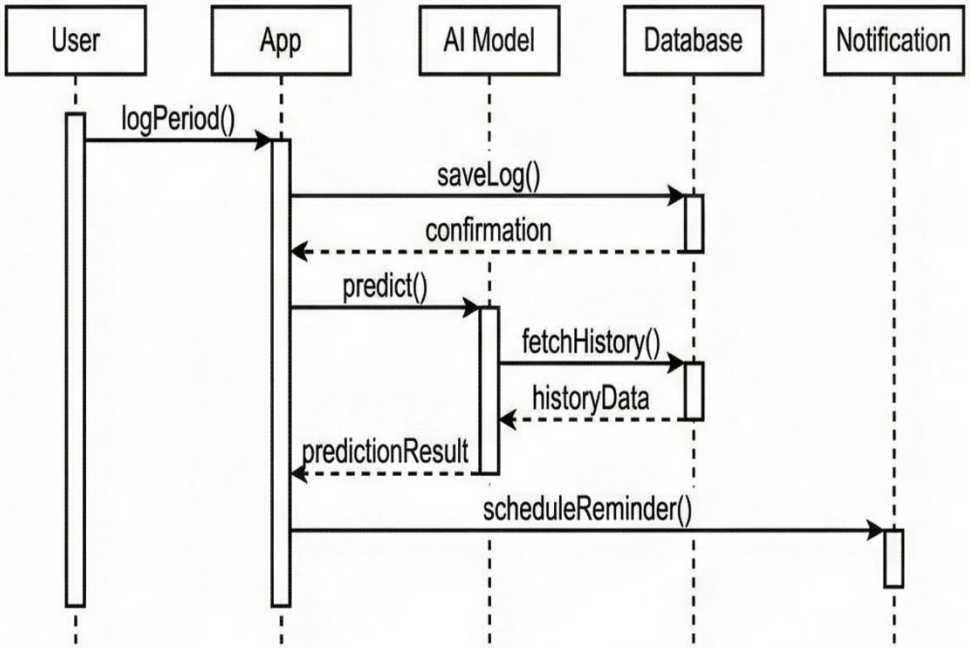


Fig.2:Sequence Diagram

Period Logging and Data Persistence: The process begins when the User sends the logPeriod() message to the App, initiating the entire sequence. The App's primary and immediate responsibility is to ensure this new data is saved; thus, it sends a saveLog() message to the Database. The Database confirms the successful storage of the new period entry by replying with a confirmation message back to the App. This establishes the single source of truth for the new entry before any subsequent operations occur.

AI-Driven Prediction and History Retrieval: Following the data save, the App leverages intelligence by sending a predict() request to the AI Model. The AI Model, needing context for an accurate forecast, then initiates a call to the Database via fetchHistory(), requesting all relevant past cycle data for the user. Once the Database returns the necessary historyData, the AI Model performs its calculations. It then returns the calculated predictionResult (e.g., the date of the next predicted period or fertile window) back to the App.

Scheduling the Reminder: In the final stage, the App uses the newly generated predictionResult to proactively engage the user. It sends a scheduleReminder() message to the Notification service. This action sets up a timely alert or push notification, ensuring the user is reminded about their upcoming period based on the AI's prediction, completing the useful cycle of logging, calculating, and notifying.

6. PARTIAL IMPLEMENTATION

Phase-I Modules Completed:

In Phase–I, a basic working prototype of the **AI Integrated Period Tracker** was developed. Initial modules such as period logging, symptom entry, and cycle history storage were implemented. The collected data was cleaned, preprocessed, and transformed into usable features for AI analysis. A simple prediction model (LSTM/Regression) was integrated to generate the first version of cycle predictions. Basic UI screens were created to display user inputs and predicted period dates. This partial implementation demonstrates the core idea of the system and forms the foundation for the advanced development in Phase–II.

Login Page:

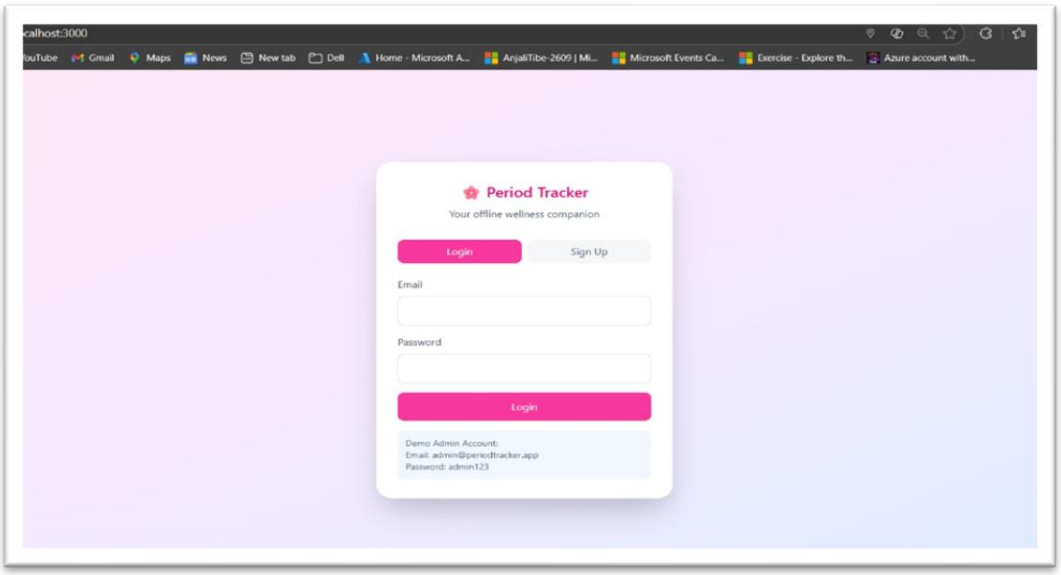


Fig.3 Login Page

Home Page:

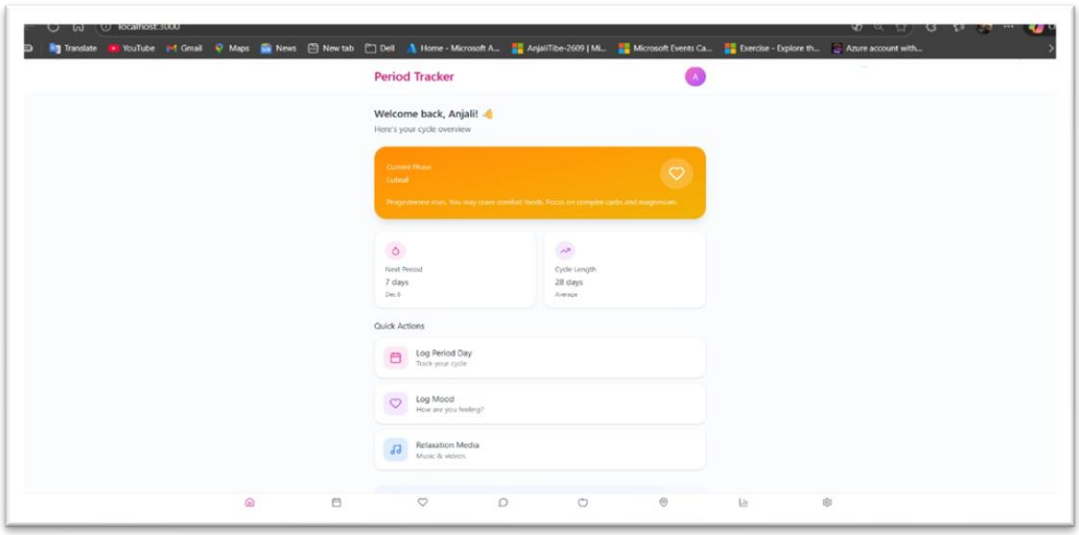


Fig.4 Home Page

Calender:

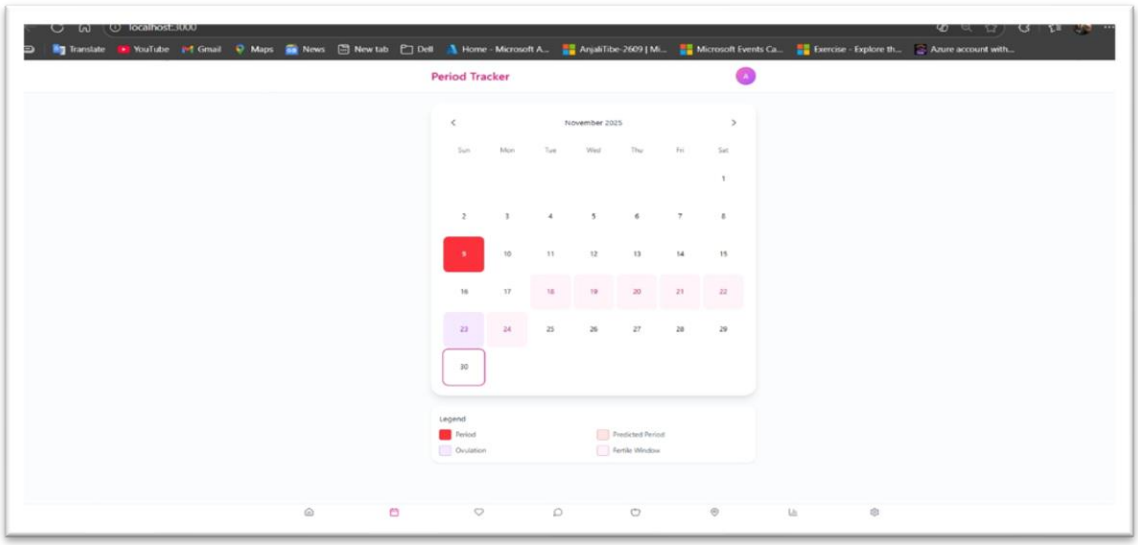


Fig.5 Calendar

Chatbot:

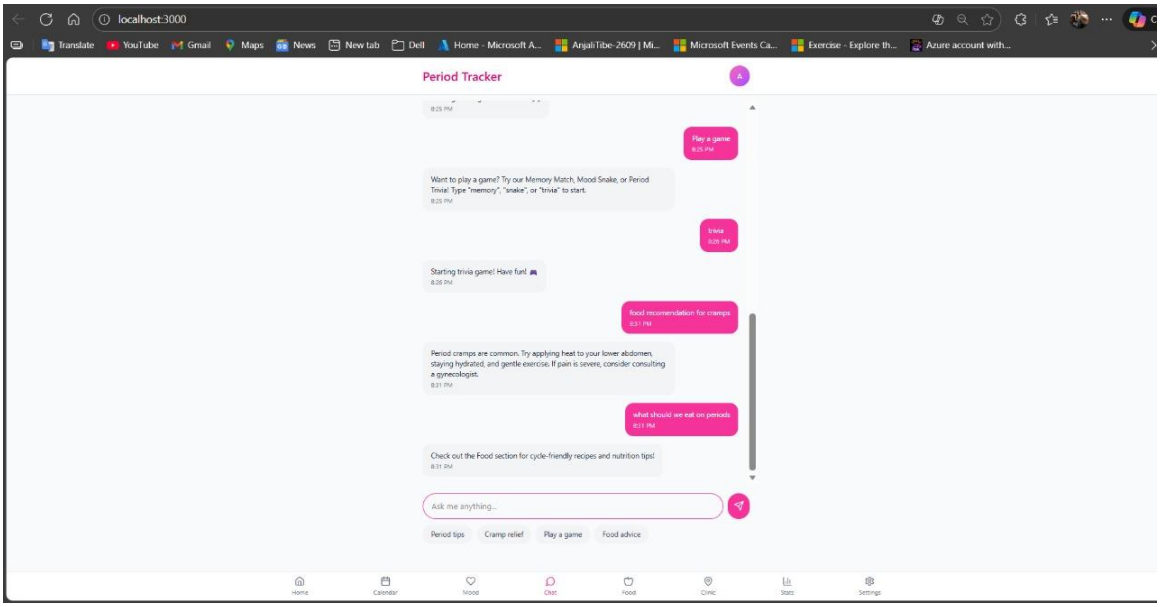


Fig.6 Chatbot

7.WORK BREAKDOWN STRUCTURE (WBS)

TIMELINE :

Phase	Task Description	Start Date	End Date	Status
Phase I	Topic Selection & Synopsis Preparation	Aug 1, 2025	Aug 15, 2025	Completed
	Literature Review & Identification of Gaps	Aug 16, 2025	Aug 31, 2025	Completed
	Data Collection (Cycle Logs & Symptoms)	Sept 1, 2025	Sept 10, 2025	Completed
	Data Cleaning & Preprocessing	Sept 11, 2025	Sept 30, 2025	Completed
	Feature Extraction (Cycle Length, Patterns)	Oct 1, 2025	Oct 20, 2025	Completed
	Model Setup & Initial Training	Oct 21, 2025	Nov 10, 2025	Completed
	Phase-I Prototype & Documentation	Nov 11, 2025	Nov 30, 2025	Completed
Phase II	App Framework Setup (Flutter/Firebase)	Jan 1, 2026	Jan 31, 2026	Pending
	AI Model Integration & Prediction Logic	Feb 1, 2026	Feb 28, 2026	Pending
	UI/UX Design & User Dashboard	Mar 1, 2026	Mar 20, 2026	Pending
	System Testing & Performance Validation	Mar 21, 2026	Apr 10, 2026	Pending
	Final Report Preparation & Submission	Apr 11, 2026	Apr 30, 2026	Pending
	13. Final Presentation & Project Demonstration	May 1, 2026	May 15, 2026	Pending

Table no.3: Timeline

GANTT CHART:

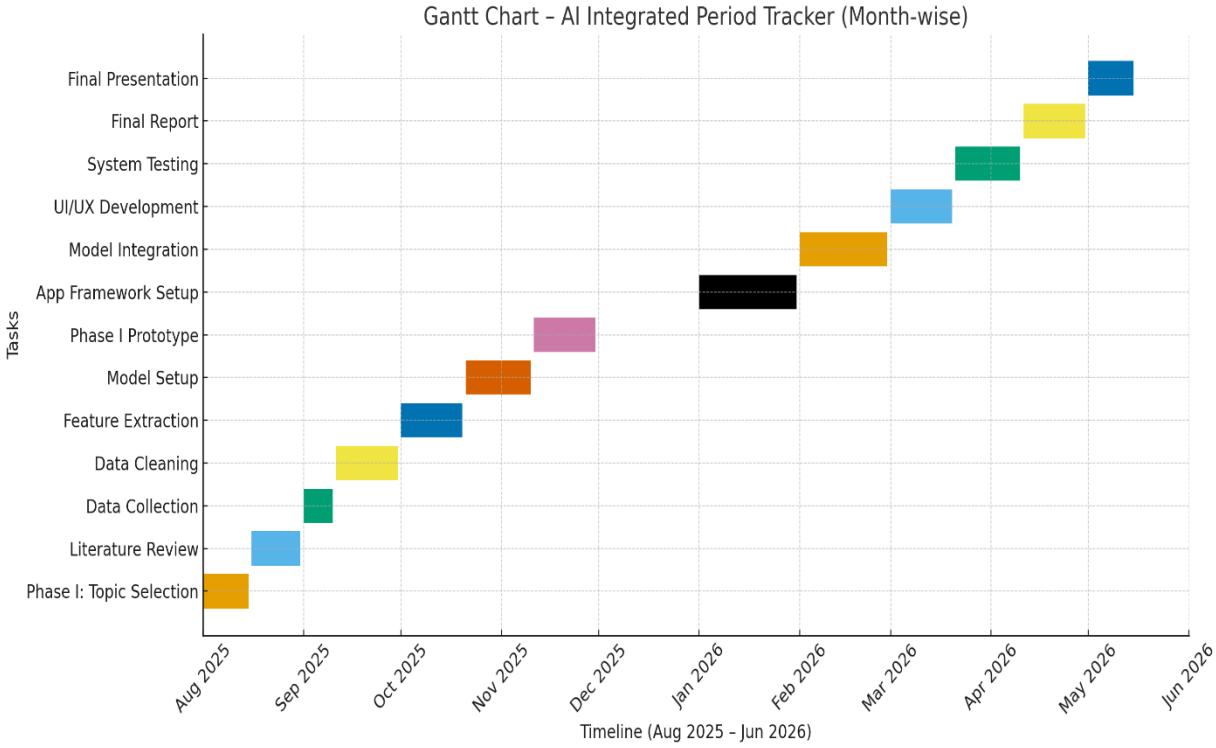


Fig.7: Project Schedule and Phase-Wise Timeline Overview

8.CONCLUSION – PHASE 1

In Phase 1 of the AI-integrated period tracker project, we focused on understanding the system requirements, defining the scope, and planning the overall architecture. This phase involved gathering both functional and non-functional requirements, analyzing user needs, and identifying the hardware and software necessary for implementing a robust and reliable system.

The system is designed to provide accurate period predictions, symptom tracking, and personalized health insights using AI and machine learning algorithms. It will allow users to input menstrual cycle data, track symptoms, and receive predictive notifications for upcoming periods or fertile windows. AI integration ensures smarter predictions over time, adapting to each user's unique cycle patterns.

This phase also highlighted the importance of data security and privacy, as sensitive personal health information will be collected and processed. The preliminary study of software and hardware requirements ensures the system will be scalable, high-performing, and user-friendly.

Overall, Phase 1 lays a solid foundation for the development of the AI-integrated period tracker, providing clear guidelines for implementation in the subsequent phases, and ensuring that the system aligns with user expectations and industry standards.

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